Goodyear Accounts Payable Chatbot – Solution Design & Functional Documentation

1. Purpose and Scope

This document describes the functional and high-level technical design of the Goodyear Accounts Payable (AP) Chatbot solution. The chatbot will serve as a centralized digital interface for both Goodyear internal users and external vendors, enabling invoice-related inquiries, approvals, and process guidance while ensuring strict data security and role-based access control.

2. Target Users

- Internal Goodyear users: AP teams, approvers, and other Goodyear employees.
- External users: Goodyear vendors accessing invoice and payment-related information.

3. Access & Authentication

- The chatbot interface, needs to be generated by the solution, will be delivered as a Goodyear-branded portal accessible to both internal users and external vendors.
- Authentication will be handled via Single Sign-On (SSO) for internal users and email-based authentication for vendors.
- User identity will be used to determine role, data access level, and available functionalities.

4. Role-Based Information Segregation

- AP Internal users will have access to full AP process information, Confluence knowledge articles, onboarding materials and invoice details.
- Other Goodyear Internal users will have access to approval capabilities, GR creation process and invoice information based on providing invoice and vendor number.
- External vendors will only be able to view data related strictly to their own invoices and payments.
- Data visibility will be restricted based on vendor ID, email domain, and user role.

5. Vendor Data Security & Access Control

- Vendors will only be able to access their own invoice data.
- Access will be validated using vendor number and registered email address.
- Internal invoice lookups will require invoice number and vendor name/number as an additional security measure.

6. Multilingual Support

- The chatbot will support multiple languages to accommodate Goodyear's global vendor and employee base.

- Language selection can be automatic (based on user profile or language detection) or manual.

7. Invoice Data Integration

- The chatbot will connect to Snowflake to retrieve invoice data.
- Invoice data will originate from SAP VIM (Vendor Invoice Management).
- The chatbot will support queries related to open invoices, invoice status, due dates, and payment blocks.

8. Knowledge Base Integration (Internal Only)

- The chatbot will be connected to the Confluence Accounts Payable workspace.
- This knowledge base will be used to answer process-related questions such as invoice submission rules, approval flows, and escalation paths
- This functionality will be available only to internal Goodyear employees.

9. AP Onboarding Support

- The chatbot will support onboarding for new AP joiners.
- It will guide users through a predefined onboarding process, including required trainings, documentation, and system access steps.
- Progress tracking and reminders can be supported where applicable.

10. Notifications and Reminders

- The chatbot will send automated reminders to approvers for overdue invoices.
- Proactive notifications can be sent for upcoming invoice approvals to reduce delays.
- Notifications will be delivered via Microsoft Teams and/or Outlook and will support direct interaction with the chatbot and it's functionalities (either by a link to the chatbot page or buttons configured for different actions).

11. Conversational Invoice Approval

- Internal approvers will be able to approve or reject service invoices directly within the chat interface.
- The chatbot will present invoice details, validation checks, and action buttons.
- Approval actions will be logged and synchronized back to SAP VIM.

12. Goods Receipt (GR) Creation Support

- When a Goods Receipt (GR) is missing, the chatbot can initiate a GR creation request.
- The chatbot will collect required information (PO number, service confirmation, quantities, dates).
- Based on validation rules, the chatbot will automate or trigger the GR creation process in SAP.

13. Collaboration Platform Integration

- The chatbot will be integrated primarily with Microsoft Teams (preferred) and optionally Outlook.
- Users will be able to interact with the chatbot, receive notifications, and perform approvals directly within Teams.

14. Branding and User Experience

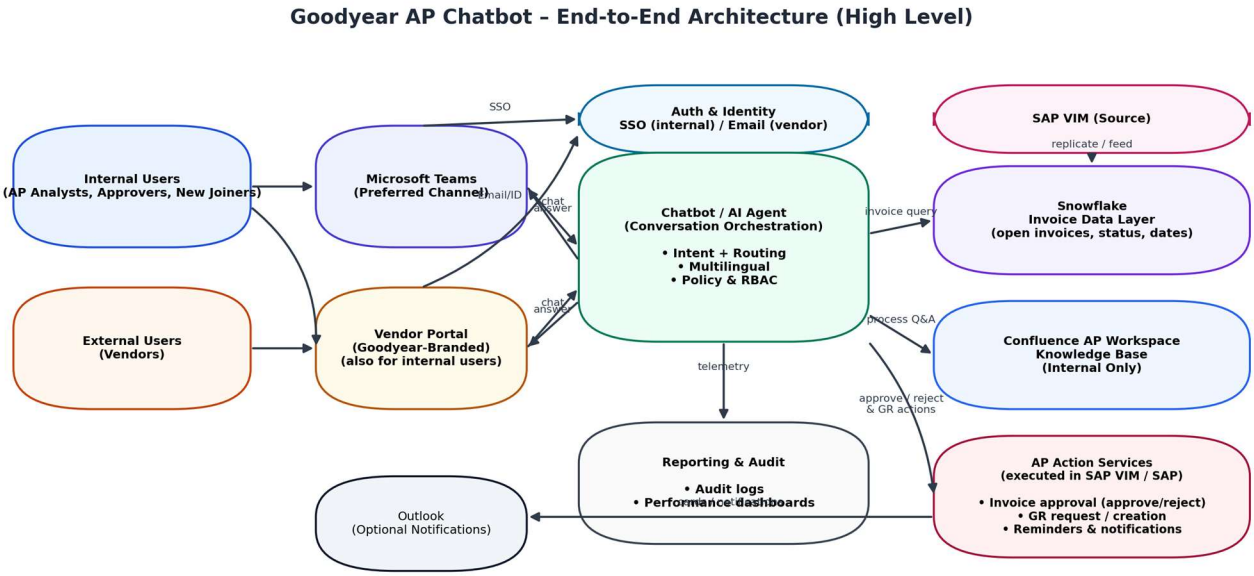
- The chatbot portal will follow Goodyear branding guidelines.
- The user interface will be intuitive, conversational, and optimized for desktop (mobile access would be a plus).

15. Security, Compliance, and Audit

- All interactions will be logged for audit and compliance purposes.
- The tool must capture and retain audit-grade logs that allow authorized teams to reconstruct and evidence chatbot activity end-to-end (who did what, when, and what the system responded).
- Data access and actions will comply with Goodyear security and data privacy standards.
- Conversation logs should avoid storing personal identifiable information (PII) unless explicitly required and approved by policy. Reference implementations note that Q&A-based chatbot logging can be configured to capture conversation and diagnostic data while avoiding collection of user identifiers such as emails/usernames/IP addresses.

16. Reporting and Monitoring capabilities

- The solution must provide comprehensive reporting and observability to support operational monitoring, compliance, and continuous improvement.
- Service Health & Reliability (Uptime of the chatbot, Error rates, Response latency, volume trends, session duration etc)
- Answer Quality & Effectiveness
    - Answer accuracy
    - Percentage of conversations resolved without human escalation.
    - Top intents/topics, unresolved questions, and knowledge gaps (to drive Confluence content improvements and training)
- Business Process Outcomes (AP Value)
    - Invoice status query volumes and trend reduction.
    - Approval cycle indicators: proactive and overdue approval counts, reminder effectiveness (e.g., approvals completed after notifications)
    - GR-related metrics: missing GR cases identified, requests initiated, completion outcomes
- Schedule Reporting and Export – ability to create custom reports and schedule recurring sending of reports to stakeholders.

17. Future Enhancements (Optional)

- Expansion to additional SAP modules.
- Advanced analytics and invoice aging insights.
- AI-driven anomaly detection for invoice issues.